

ALGEBRA

1. Solve the following inequation and write down the solution set:

$$11x - 4 < 15x + 4 \leq 13x + 14, x \in W \quad (1)$$

Represent the solution on a real number line.

2. Solve for x the quadratic equation $x^2 - 4x - 8 = 0$.
Give your answer correct to three significant figures.
3. Using the factor theorem, show that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.
Hence factorise the polynomial completely.
4. Using the Remainder Theorem find the remainders obtained when $x^3 + (kx + 8)x + k$ is divided by $x + 1$ and $x - 2$.
Hence find k if the sum of the two remainders is 1.
5. In an Arithmetic Progression (A.P.) the fourth and sixth terms are 8 and 14 respectively. Find the:
- (a) first term
 - (b) common difference
 - (c) sum of the first 20 terms.
6. The first and last term of a Geometrical Progression (G.P.) are 3 and 96 respectively. If the common ratio is 2, find:
- (a) n , the number of terms of the G.P.
 - (b) Sum of the n terms.
7. The sum of the first three terms of an Arithmetic Progression (A.P.) is 42 and the product of the first and third term is 52. Find the first term and the common difference.
8. Using properties of proportion solve for x , given

$$\frac{\sqrt{5x} + \sqrt{2x - 6}}{\sqrt{5x} - \sqrt{2x - 6}} = 4 \quad (2)$$

9. The product of two consecutive natural numbers which are multiples of 3 is equal to 810. Find the two numbers.

Trigonometry

10. Prove that:

$$(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta) = 1 \quad (3)$$

11. Simplify

$$\sin A \begin{bmatrix} \sin A & -\cos A \\ \cos A & \sin A \end{bmatrix} + \cos A \begin{bmatrix} \cos A & \sin A \\ -\sin A & \cos A \end{bmatrix}$$

12. A man observes the angle of elevation of the top of the tower to be 45° . He walks towards it in a horizontal line through its base. On covering 20m the angle of elevation changes to 60° . Find the height of the tower correct to 2 significant figures.

Geometry

13. M and N are two points on the X axis and Y axis respectively.

P (3, 2) divides the line segment MN in the ratio 2 : 3.

Find.

- (a) the coordinates of M and N
- (b) slope of the line MN

14. The vertices of a $\triangle ABC$ are $A(3, 8)$, $B(-1, 2)$ and $C(6, -6)$. Find:

- (a) Slope of BC
- (b) Equation of a line perpendicular to BC and passing through A.

15. Use ruler and compass only for answering this question.
 Draw a circle of radius 4cm. Mark the centre as O. Mark a point P outside the circle at a distance of 7cm from the centre. Construct two tangents to the circle from the external point P.
 Measure and write down the length of any one tangent.
16. Using ruler and a compass only, construct a semi-circle with diameter $BC = 7\text{cm}$.
 Locate a point A on the circumference of the semicircle such that A is equidistant from B and C. Complete the cyclic quadrilateral ABCD, such that D is equidistant from AB and BC. Measure $\angle ADC$ and write it down.
17. In the given figure AC is a tangent to the circle with center O.
 If $\angle ADB = 55^\circ$, find x and y. Give reasons for your answers.
18. In the given figure, ABCDE is a pentagon inscribed in a circle such that AC is a diameter and side $BC \parallel AE$. If $\angle BAC = 50^\circ$, find giving reasons:
- $\angle ACB$
 - $\angle EDC$
 - $\angle BEC$
- Hence prove that BE is also a diameter
19. In the given figure, $\angle PQR = \angle PST = 90^\circ$, $PQ = 5\text{cm}$, and $PS = 2\text{cm}$.
- Prove that $\triangle PQR \sim \triangle PST$.
 - Find Area of $\triangle PQR$: Area of quadrilateral SRQT.

Mensuration

20. A solid metallic sphere of radius 6cm is melted and made into a solid cylinder of height 32cm. Find the:
- radius of the cylinder
 - curved surface area of the cylinder

21. A hemispherical and a conical hole is scooped out of a solid wooden cylinder. Find the volume of the remaining solid where the measurements are as follows:
The height of the solid cylinder is 7cm, radius of each of hemisphere, cone and cylinder is 3cm. Height of cone is 3cm.
Give your answer correct to the nearest whole number. Take $\pi = \frac{22}{7}$.

Statistics and Probability

22. In a class of 40 students ,marks obtained by the students in a class test (out of 10) are given below:

Marks	1	2	3	4	5	6	7	8	9	10
Number of students	1	2	3	3	6	10	5	4	3	3

Calculate the following for the given distribution:

- (a) Median
 - (b) Mode
23. The data on the number of patients attending a hospital in a month are given below.
Find the average (*mean*) number of patients attending the hospital in a month by using the shortcut method.
Take the assumed mean as 45. Give your answer correct to 2 decimal places.
- | Number of patients | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 |
|--------------------|-------|-------|-------|-------|-------|-------|
| Number of Days | 5 | 2 | 7 | 9 | 2 | 5 |
24. There are 25 discs numbered 1 to 25. They are put in a closed box and shaken thoroughly. A disc is drawn at random from the box.
Find the probability that the number on the disc is:
- (a) an odd number
 - (b) divisible 2 and 3 both.
 - (c) a number is less than 16.
25. Use graph for this question.
The marks obtained by 120 students in a English test are given below: Draw the ogive and hence, estimate:

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No.of students	5	9	16	22	26	18	11	6	4	3

- (a) the median marks.
- (b) the number of students who did not pass the pass percentage was 50.
- (c) the upper quartile marks.

Commercial Mathematics

26. A man invests ₹ 4500 in shares of a company which is paying 7.5% dividend. If ₹ 100 shares are available at a discount of 10%. Find:
- (a) Number of shares he purchases.
 - (b) His annual income.
27. Sachin invests ₹ 8500 in 10%, ₹ 100 shares at ₹ 170. He sells the shares when the price of each share rises by ₹ 30. He invests the proceeds in 12% ₹ 100 shares at ₹ 125. Find:
- (a) the sale proceeds.
 - (b) the number of ₹ 125 shares he buys.
 - (c) the change in his annual income.
28. Rekha opened a recurring deposit account for 20 months. The rate of interest is 9% per annum and Rekha receives ₹ 441 as interest at the time of maturity. Find the amount Rekha deposited each month.

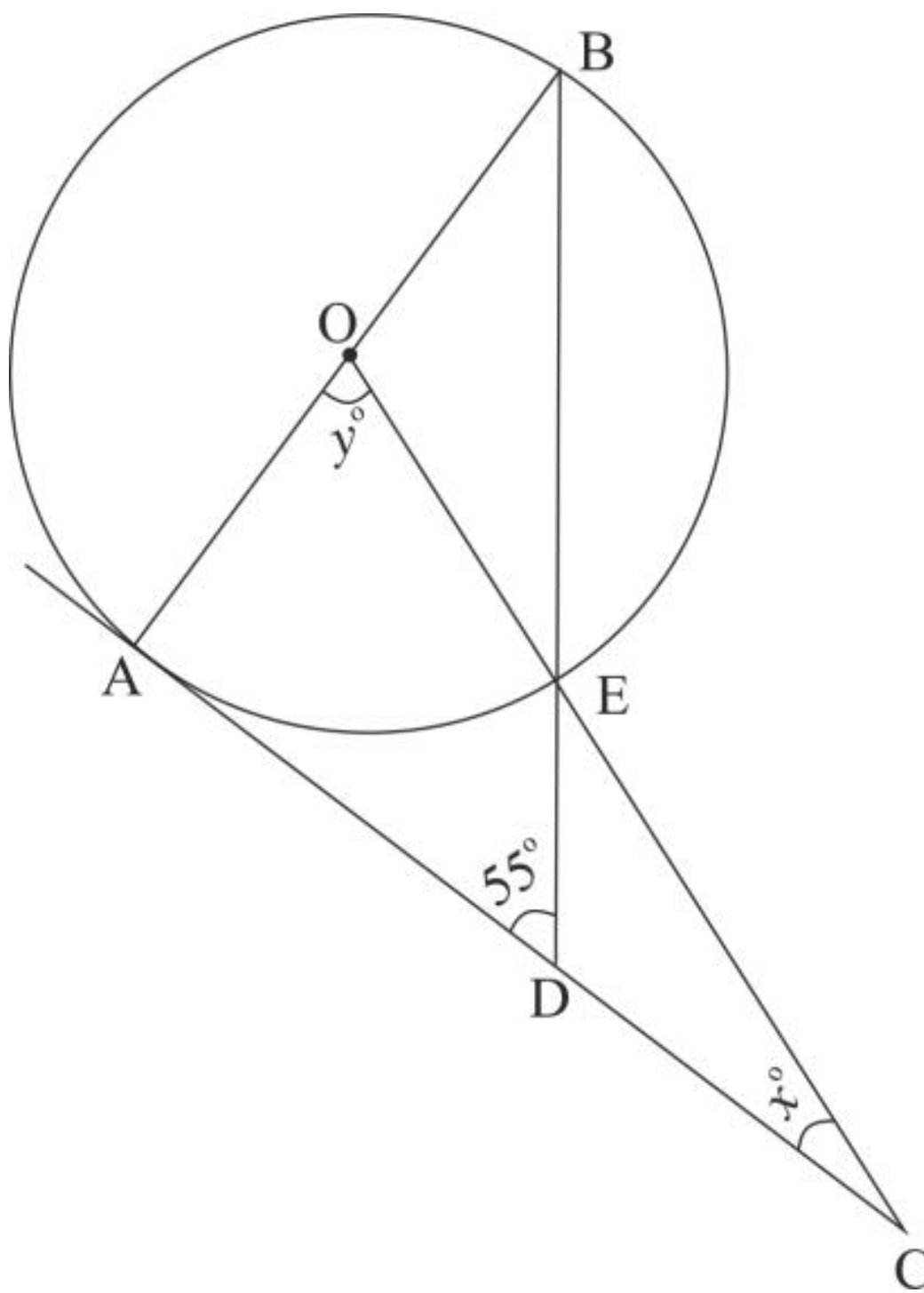


Figure 1

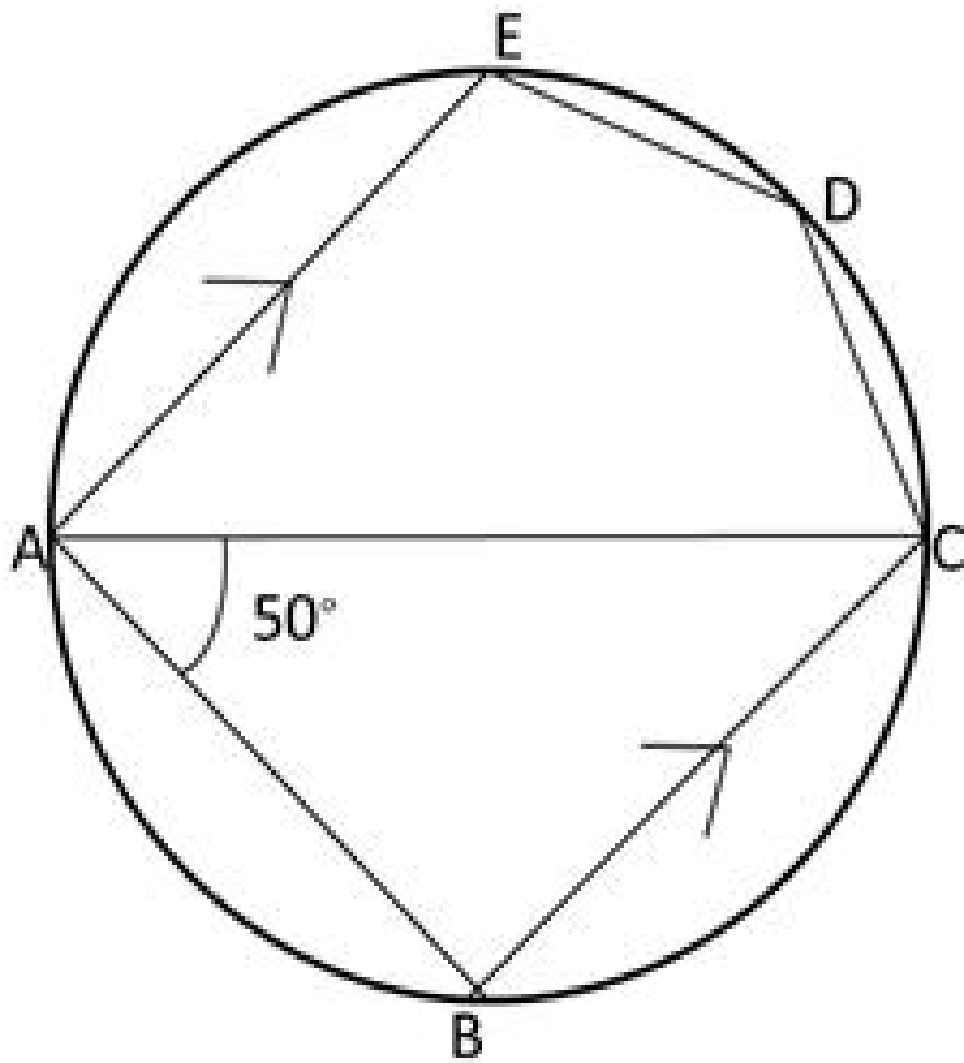


Figure 2

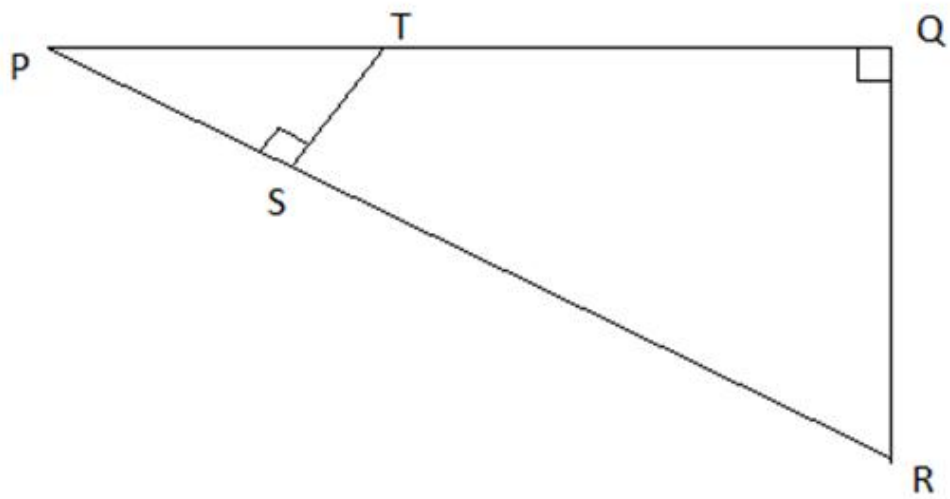


Figure 3

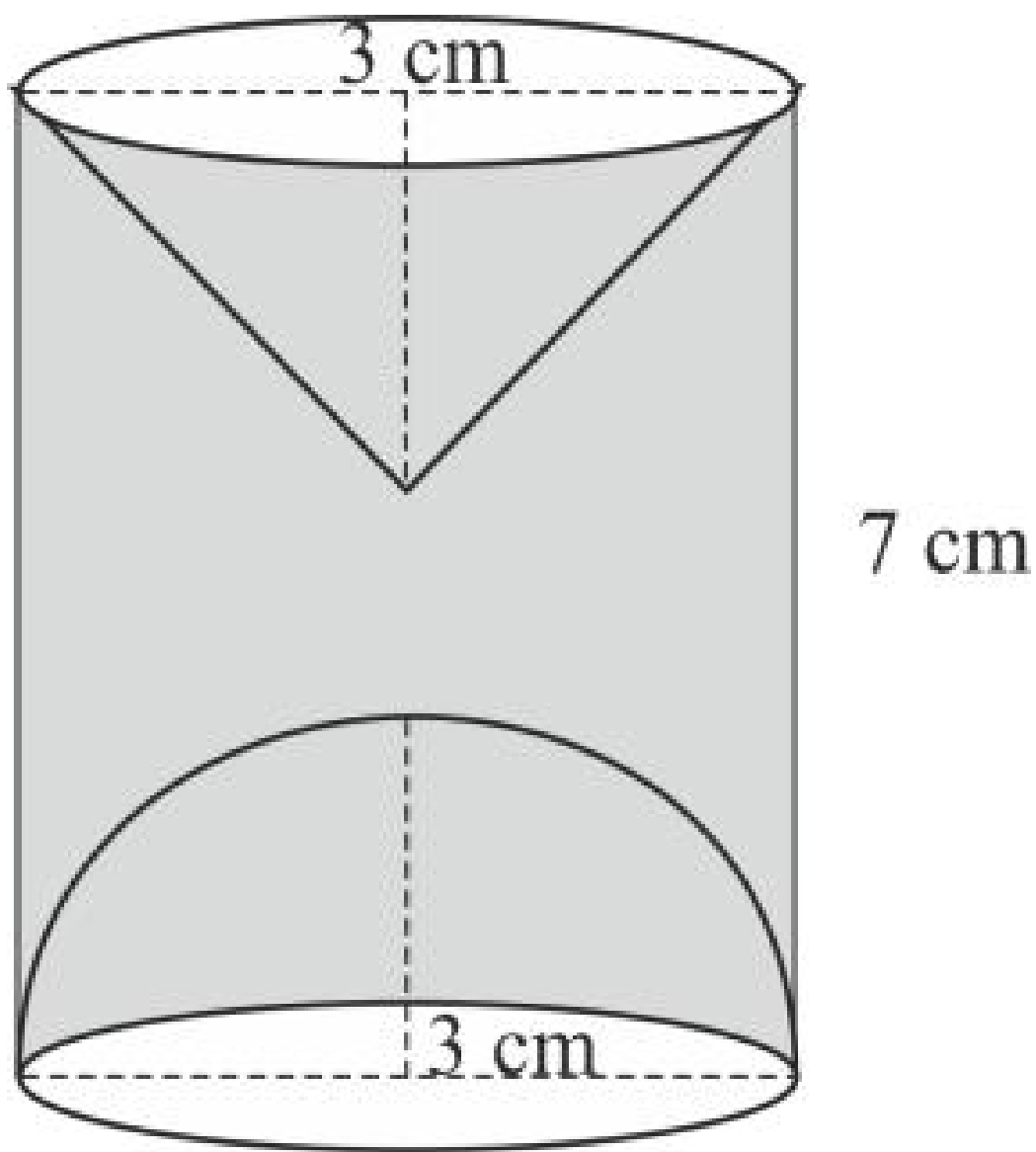


Figure 4